SD-TSIA205

Exercise 3

Regression

Let $f : \mathbb{R} \to \mathbb{R}$ be an unknown function. Suppose we observe noisy values of f :

$$Y_n = f(X_n) + \epsilon_n,$$

where

— $(X_n, \epsilon_n)_{n=1,\dots,N} \subset \mathbb{R} \times \mathbb{R}$ is an i.i.d. collection of random variables such that $\epsilon \perp X$, $\mathbb{E}[\epsilon] = 0$, $\mathbb{E}[\epsilon^2] = \sigma^2$ and $\mathbb{E}[f(X)^2] < \infty$.

— To estimate the function f, we shall use a sequence of functions $(h_k)_{k\geq 1}$ where each $h_k : \mathbb{R} \to \mathbb{R}$ and $G = \mathbb{E}[h(X)h(X)^T]$ is invertible.

The procedure is as follows. We choose K, we set $h(x) = (h_1(x), \dots h_K(x))^T$ and then compute

$$\theta_{K,N} \in \arg\min_{\theta\in\mathbb{R}^K} \sum_{n=1}^{N} (Y_n - h(X_n)^T\theta)^2.$$

The estimate of f is given by $f_{K,N}(x) = h(x)^T\theta_{K,N}$. Define the risk $R(\theta) = \mathbb{E}[(f(X) - h(X)^T\theta)^2]$. The aim is to study the $R(\theta_{K,N})$.

1. Let $\theta_K^* \in \arg\min_{\theta\in\mathbb{R}^K} R(\theta)$. Give the normal equations satisfied by θ_K^* and deduce an expression for θ_K^*.
2. Give the expression for $\theta_{K,N}$.
3. Show that the estimated function $f_{K,N}$ is invariant by any linear invertible transform on the set of functions h, i.e. h is replaced by Ah where $A \in \mathbb{R}^{K\times K}$ is invertible. In the following we shall assume that $G = I_K$.
4. Show that for any $\theta \in \mathbb{R}^K$, $R(\theta) = R(\theta_K^*) + \|\theta - \theta_K^*\|^2$ where the norm $\|\cdot\|$ should be specified.
5. From now on, we suppose that the smallest eigenvalue of $N^{-1}\sum_{n=1}^{N} h(X_n)h(X_n)^T$ is lower bounded by $\lambda > 0$. Show that $\|\theta_{K,N} - \theta_K^*\| \leq \lambda^{-1/2}\|N^{-1}\sum_{n=1}^{N}\xi_n h(X_n)\|$ where $\xi_n \neq \epsilon_n$ should be specified.
6. Show that $N^{-1}\sum_{n=1}^{N}\xi_n h(X_n) \to 0$ almost surely (Hint : use the Cauchy-Schwarz inequality and the strong law of large numbers).
7. Suppose that $f \in \mathrm{span}((h_k)_{k\geq 1})$. Conclude that choosing K large enough, $\limsup_{N\to\infty} R(\theta_{K,N})$ can be made arbitrarily small.